UNIT V

TOUGH MISSING-BLOCK MCQ QUIZ

Hierarchical RL • MAXQ • Meta-RL • MARL • POMDP • Ethics

10 SCENARIO-BASED QUESTIONS

Fill the missing concept by selecting the best option.

Q1 • HRL — Temporal Abstraction

A robot must deliver an item through several rooms. A high-level policy selects the current room as a sub-goal, while a lower-level policy performs navigation actions repeatedly until that sub-goal is completed. This design primarily demonstrates □□□□.

□□□□

Select the option that correctly fills the missing concept.

A. reward shaping

B. temporal abstraction

C. experience replay

D. policy distillation

Q2 • HRL — Hierarchy Design

An RL system has a mission policy that changes only when a major objective is completed, while a skill policy changes actions at every time step. The key advantage is that the upper policy operates at a □□□□ time scale than the lower policy.

□□□□

Select the option that correctly fills the missing concept.

A. faster

B. slower

C. randomized

D. continuous

Q3 • HAM — Structured Exploration

In a warehouse task, the designer knows that an agent must first locate an item, then pick it up, and finally deliver it. The task structure is encoded before learning, while some choices inside the structure are learned. This most directly describes a □□□□.

□□□□

Select the option that correctly fills the missing concept.

A. Hierarchical Abstract Machine

B. Markov reward process

C. multi-armed bandit

D. policy gradient baseline

Q4 • MAXQ — Value Decomposition

A navigation task is decomposed into Navigate-To-Room, Pick-Object, and Deliver-Object subtasks. Instead of learning one monolithic value function, the system estimates values associated with the subtasks and their parent task. This is characteristic of □□□□.

□□□□

Select the option that correctly fills the missing concept.

A. MAXQ decomposition

B. Monte Carlo tree search

C. Q-learning without states

D. belief-state filtering

Q5 • Meta-RL — Fast Adaptation

During training, an RL agent encounters hundreds of related traffic-control tasks with different traffic patterns. At deployment, it receives only a small amount of experience from a new pattern and adapts rapidly. The training objective is best described as □□□□.

□□□□

Select the option that correctly fills the missing concept.

A. learning a transferable learning strategy

B. maximizing a fixed single-task policy

C. removing task variability completely

D. replacing rewards with supervised labels

Q6 • MARL — Non-Stationarity

Two autonomous vehicles learn simultaneously. Vehicle A updates its policy, changing the behavior that Vehicle B observes, even though the road environment itself has not changed. From Vehicle B's learning perspective, the environment becomes effectively □□□□.

□□□□

Select the option that correctly fills the missing concept.

A. stationary

B. non-stationary

C. fully observable

D. deterministic

Q7 • MARL — Credit Assignment

Four agents cooperate to complete a task, but only one shared reward is received after completion. The main difficulty is determining which agent's actions contributed to the final outcome. This challenge is called □□□□.

□□□□

Select the option that correctly fills the missing concept.

A. credit assignment

B. temporal abstraction

C. belief updating

D. domain randomization

Q8 • POMDP — Belief State

A robot cannot directly observe whether a corridor is occupied. It receives noisy sensor readings and maintains probabilities over possible hidden states before selecting an action. The probability distribution used for decision-making is its □□□□.

□□□□

Select the option that correctly fills the missing concept.

A. belief state

B. reward function

C. action-value table

D. termination condition

Q9 • Ethics — Reward Hacking

An RL system is rewarded for reducing delivery time. It discovers a behavior that minimizes the measured time but repeatedly violates an important safety constraint that was not included in the reward. This is a classic example of □□□□.

□□□□

Select the option that correctly fills the missing concept.

A. reward hacking

B. temporal abstraction

C. centralized training

D. partial observability

Q10 • Integrated Real-World RL

A smart traffic controller is first trained in simulation, evaluated statistically, checked against safety constraints, and only then considered for deployment. This workflow emphasizes □□□□ before real-world operation.

□□□□

Select the option that correctly fills the missing concept.

A. validation and safety evaluation

B. unrestricted exploration

C. reward removal

D. single-agent randomization